

Case Reference: CASE-TXN-1638CE92
Transaction ID: TXN-1638CE92
Timestamp: 2026-08-16 15:21:29 UTC
Amount: **\$1,850.00 USD**
Action Taken: AUTO_BLOCK

RISK LEVEL: CRITICAL
96.6/100

Status: Transaction restricted; dossier preserved
for compliance and audit trail.

1. Multi-Factor Risk Model Diagnostics

Diagnostic Factor	Raw Metric	Factor Weight	Normalized Impact
Supervised Machine Learning (XGBoost)	99.3% Fraud Probability	65%	64.6 / 65.0
Unsupervised Anomaly (Isolation Forest)	96.0 Outlier Index	25%	24.0 / 25.0
Domain Heuristics & Rules	2 Rule(s) Triggered	10%	5.0 / 10.0
CALIBRATED COMPOSITE RISK SCORE	CRITICAL SEVERITY	100%	96.6 / 100.0

2. Explainable AI (SHAP) Forensic Feature Attribution

Primary Behavioral Triggers Identified:

- Latent identity / card-use deviation (V14 component)
- Transaction amount pattern (\$1,850.00)
- Multi-vector behavioral anomaly score

Feature Name	Observed Value	SHAP Attribution (Delta)	Risk Direction
V14	-5.24	+0.3640	Elevates Fraud Probability
V12	-3.88	+0.2810	Elevates Fraud Probability
Amount_Deviation_Z	2.45	+0.1650	Elevates Fraud Probability

3. Fraud Analyst Investigation Notes & Audit Signature

Investigator: Lead Fraud Analyst #8142

Findings: Explorer review: Action recommendation AUTO_BLOCK

Digital Audit Checksum (SHA-256):

f4232c5832d27efd8bfed74f4c86a718a3ede99c2d88d104dbbae3d05eff256c

Verified Fraud Risk
Engine

COMPLIANCE CERTIFIED